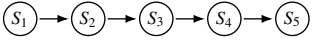
**(a) Directed chain**

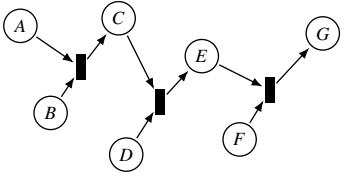



one-way conversion

**(b) Reversible chain**

forward and reverse steps

**(c) Random sparse bimolecular**




two substrates per reaction; sparse connectivity